

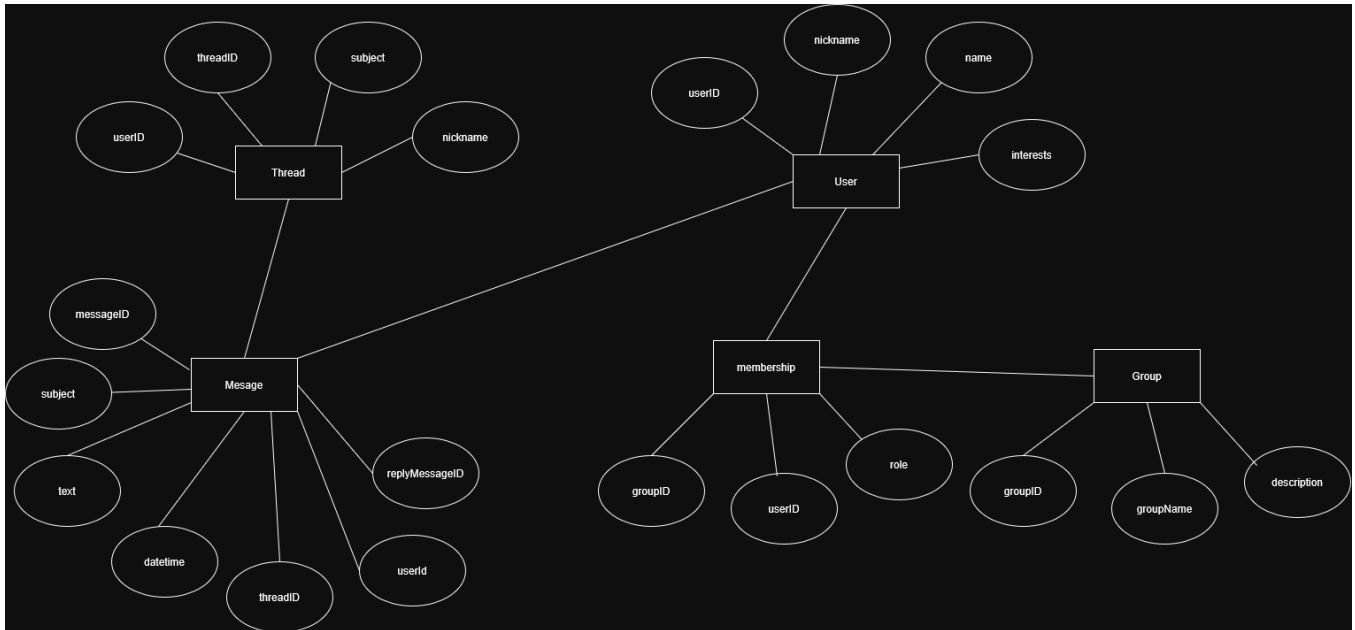
Advanced Database Models Exercise 3

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Question 1

Q1 Part (a)



Q1 Part (b)

Yes, subject is redundant.

Normal Form	Conforms	Reason
1NF	✓	
2NF	✗	Because in subject in thread table can be same for multiple threadID values, therefore not dependent on the PK
3NF	✗	Because the schema is not in 2NF

🔍 Question

If subject has duplicate values for different threadID then does it mean that 2NF is not satisfied?

Question 2

Q2 1

```
π(messageID, subject, text, datetime, threadID, userID, replyMessageID) σ(nickname='elwood')(message ⋈ (userID = userID) user)
```

Q2 2

```
π(userID, nickname, name) σ(groupname='zanzibar')(group ⋈ (groupID = groupID) membership ⋈ user (userID = userID))
```

Q2 3

```
 $\pi(\text{messageID}, \text{subject}, \text{text}, \text{datetime}, \text{threadID}, \text{userID}, \text{replyMessageID}) \sigma(\text{subject} = \text{'New Neil Young album out!'})(\text{thread} \bowtie (\text{threadID} = \text{threadID}) \text{message})$ 
```

Q2 4

```
 $\pi(\text{messageID}, \text{subject}, \text{text}, \text{datetime}, \text{threadID}, \text{userID}, \text{replyMessageID}) \sigma(\text{subject} = \text{'New Ron Sexsmith album out!'})(\text{thread} \bowtie (\text{threadID} = \text{threadID}) \text{message})$ 
```

Note

In relational algebra, the order of tuples in a relation has no meaning or importance, hence there is no established way to write it/include it

Q2 5

```
 $\pi(\text{threadID}, \text{subject}, \text{userID}, \text{nickname}) \sigma(\text{groupname} = \text{'kiddies'})(\text{thread} \bowtie (\text{threadID} = \text{threadID}) \bowtie \text{membership} \bowtie (\text{groupID} = \text{groupID}) \text{group})````$ 
```

Q2 6

Q2 7

```
 $\pi(\text{userID}, \text{nickname}, \text{name}, \text{interests}) \sigma(\text{messageID} = \text{NULL})(\text{message} \bowtie (\text{userID} = \text{userID}) \text{user})$ 
```

OR possibly,

```
 $\pi(\text{userID})(\text{user}) \setminus \pi(\text{userID})(\text{message})$ 
```

Note

Anti-join is used to find non-existence in relational algebra, instead of `=NULL` approach

Q2 8